

LAB SETUP REPORT

Foundation & Environment Setup

Practical Cybersecurity Laboratory

Attacker Machine: Kali Linux

Target Machine: Metasploitable2 / DVWA

Virtualization Platform: VMware

Network: NAT + Host-only

Prepared for academic practical work

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1. Introduction and Objective

This project establishes a controlled cybersecurity laboratory for learning cybersecurity, networking, Linux administration, cryptography, and security-tool fundamentals. Kali Linux is used as the security-testing workstation, while Metasploitable2 and DVWA provide intentionally vulnerable targets for authorized practice.

Objective

To build strong fundamentals in cybersecurity, networking, cryptography, and practical security testing by creating an isolated virtual lab and completing basic hands-on exercises.

- Understand the CIA Triad, common threat types, and attack vectors.
- Configure a private virtual lab using VMware, Kali Linux, and Metasploitable2/DVWA.
- Practice Linux filesystem navigation, permissions, package management, and networking commands.
- Study OSI/TCP-IP, DNS, HTTP/HTTPS, IP addressing, subnetting, and NAT.
- Perform basic encryption, decryption, hashing, certificate, and TLS exercises with OpenSSL.
- Become familiar with Nmap, Wireshark, Burp Suite, and Netcat.

2. Cybersecurity Basics

2.1 CIA Triad

Principle	Meaning	Example
Confidentiality	Only authorized users can access information.	Passwords, access control, encryption
Integrity	Information remains accurate and is not modified without authorization.	Hashing, digital signatures

Availability	Systems and information remain accessible when needed.	Backups, redundancy, DDoS protection
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2.2 Threat Types

Threat	Description
Phishing	Fraudulent messages or websites intended to trick users into revealing information.
Malware	Malicious software such as viruses, worms, trojans, and spyware.
DDoS	An attack intended to make a service unavailable by overwhelming it with traffic or requests.
SQL Injection	Malicious input that interferes with an application's database query.
Brute Force	Repeatedly trying credentials until a valid password is found.
Ransomware	Malware that blocks access to data, commonly by encrypting files and demanding payment.

2.3 Attack Vectors

Social Engineering: Manipulating people into revealing information or performing unsafe actions.

Wireless Attacks: Attacks targeting wireless communication, authentication, or configuration.

Insider Threats: Intentional or accidental misuse of legitimate access by an authorized user.

3. Lab Environment Setup

The lab was configured in VMware using Kali Linux as the attacker/security workstation and Metasploitable2 as the vulnerable target. Kali was configured with NAT for Internet connectivity and a Host-only interface for the isolated target network. Metasploitable2 was connected to the Host-only network.

3.1 Virtual Machine Network Configuration

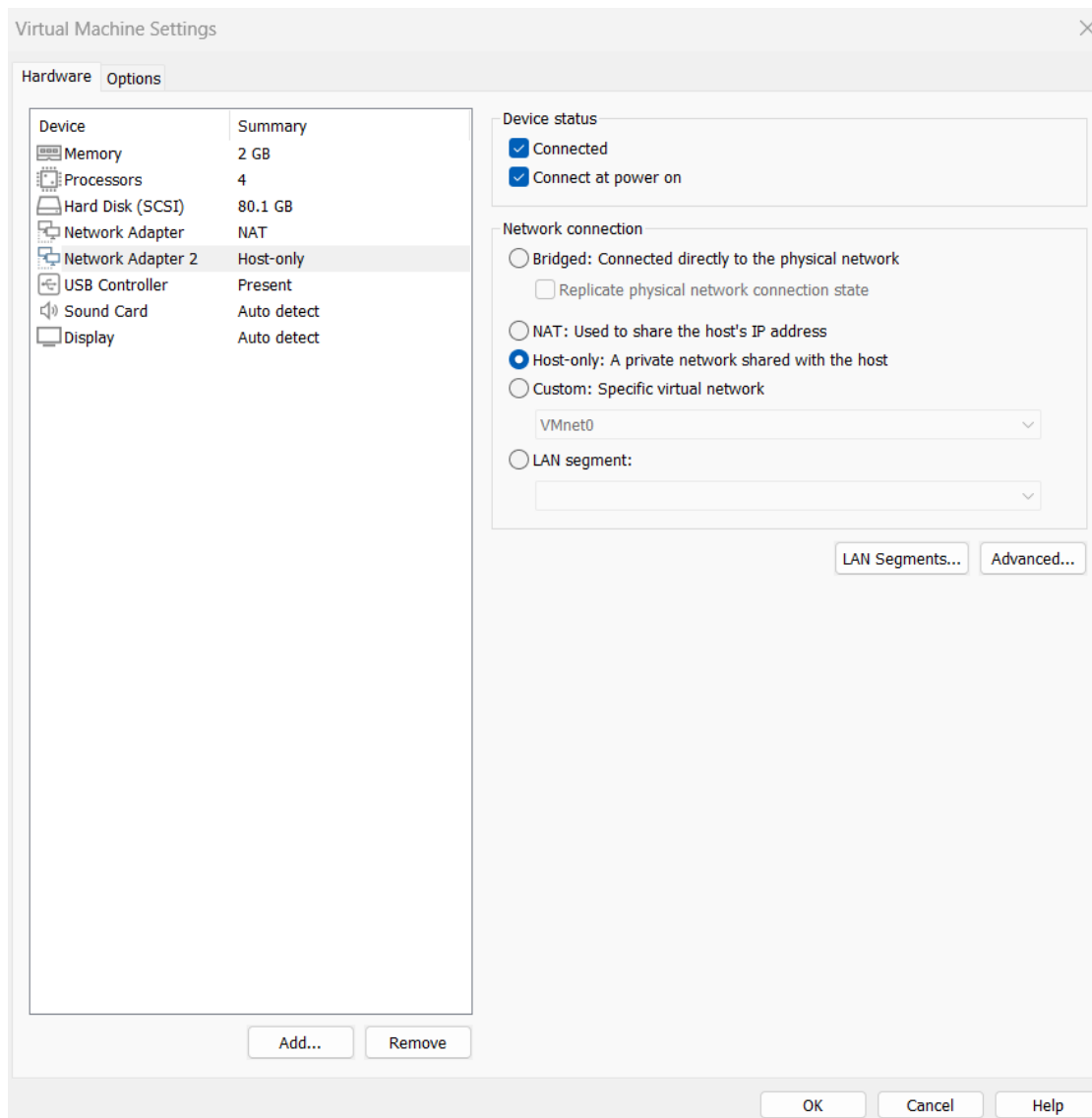


Figure 1. Kali Linux Host-only adapter configuration.

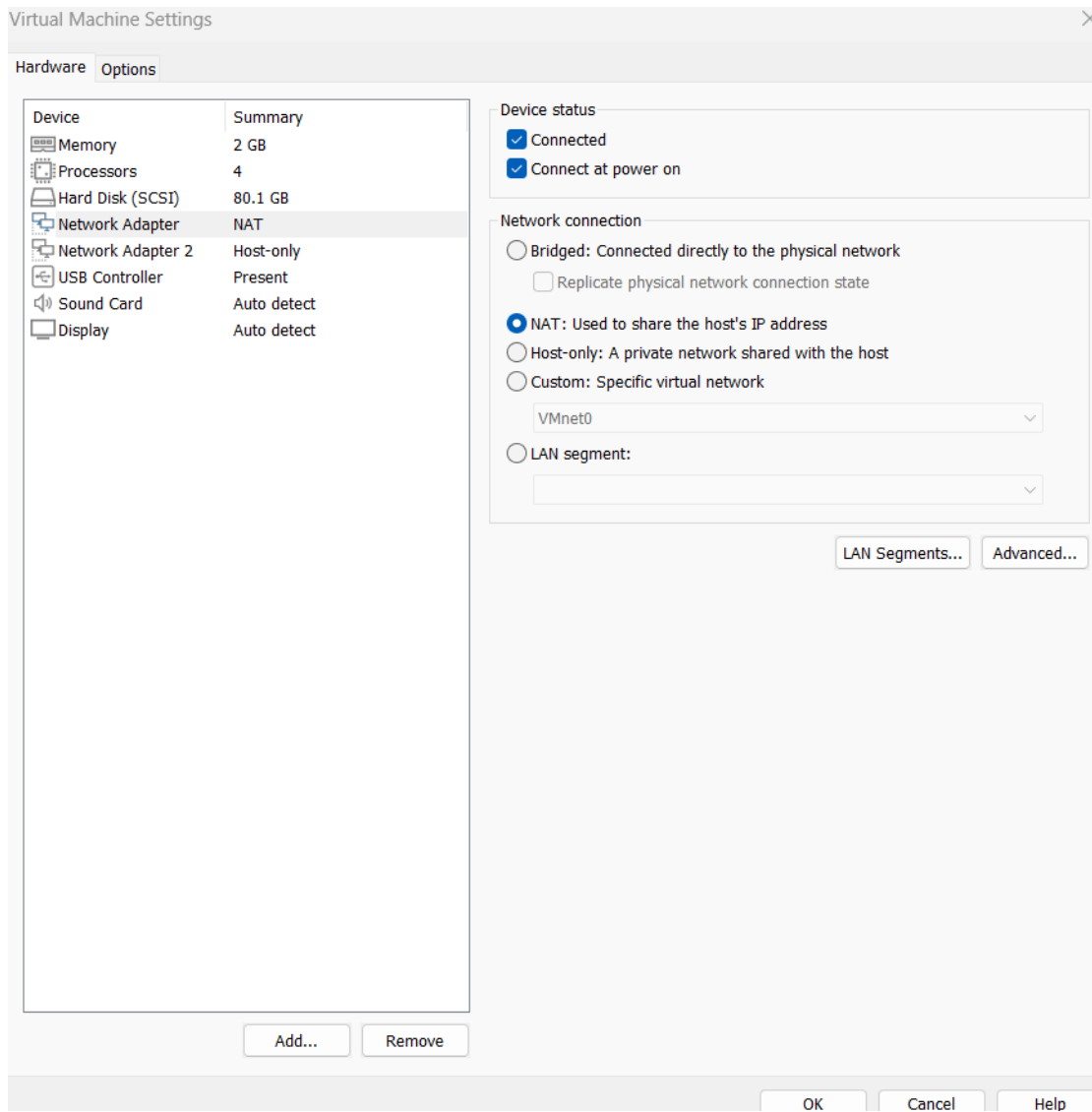


Figure 2. Kali Linux NAT adapter configuration.

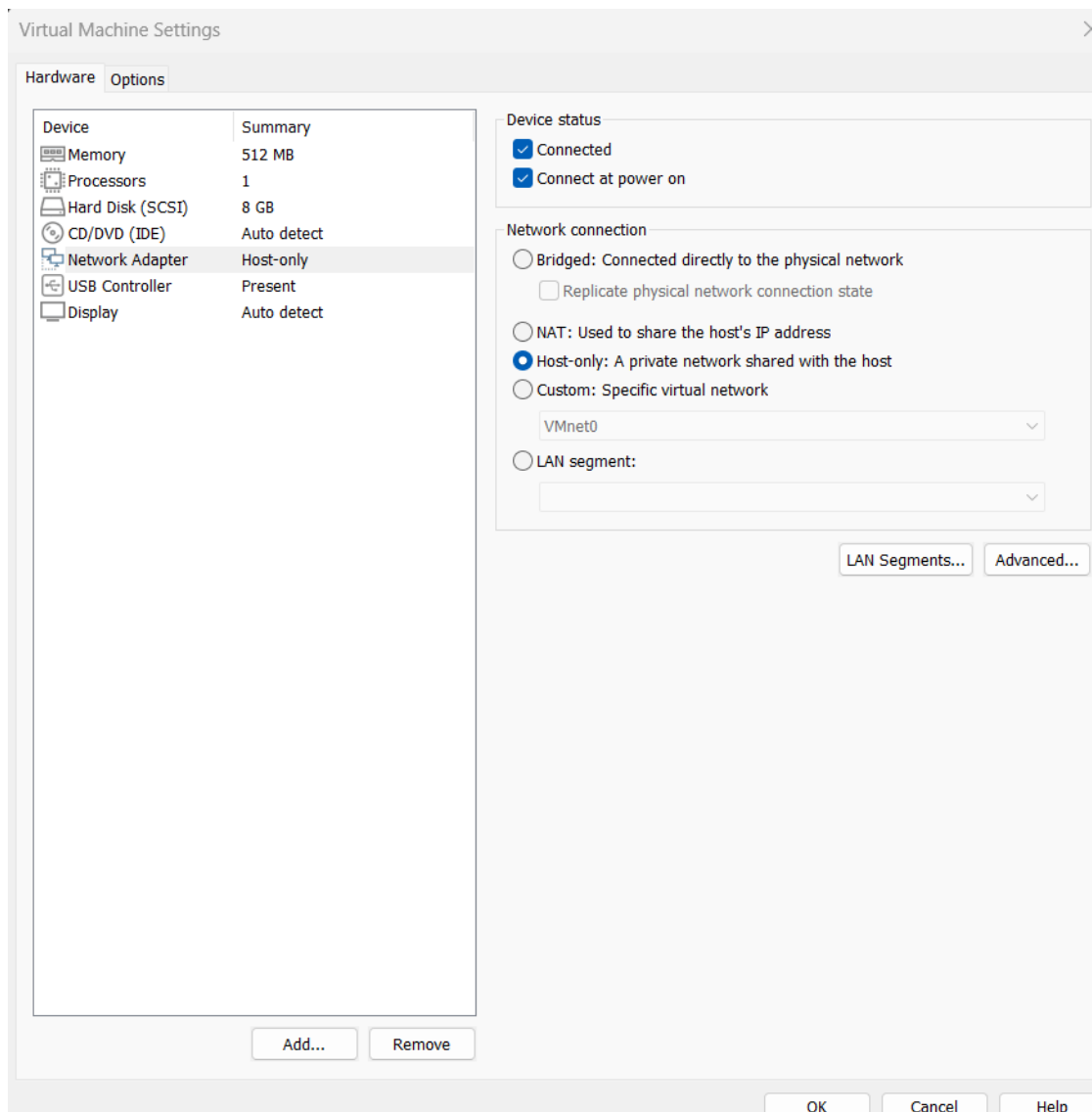


Figure 3. Metasploitable2 Host-only adapter configuration.

3.2 IP Configuration and Connectivity

The lab used the 192.168.133.0/24 Host-only subnet. Kali used 192.168.133.129 on its Host-only interface and Metasploitable2 used 192.168.133.128. A ping test verified communication between the two machines.

```
(kali㉿kali)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:03:0e:de brd ff:ff:ff:ff:ff:ff
    inet 192.168.241.130/24 brd 192.168.241.255 scope global dynamic noprefixroute eth0
        valid_lft 1466sec preferred_lft 1466sec
    inet6 fe80::1df2:289b:c3ad:8f58/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:03:0e:e8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.133.129/24 brd 192.168.133.255 scope global dynamic noprefixroute eth1
        valid_lft 1466sec preferred_lft 1466sec
    inet6 fe80::5e8:c61d:4ad0:1008/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
(kali㉿kali)-[~]
$
```

Figure 4. Kali Linux IP configuration.

```
(kali㉿kali)-[~]
$ ping 192.168.133.128
PING 192.168.133.128 (192.168.133.128) 56(84) bytes of data:
64 bytes from 192.168.133.128: icmp_seq=1 ttl=64 time=0.549 ms
64 bytes from 192.168.133.128: icmp_seq=2 ttl=64 time=0.877 ms
64 bytes from 192.168.133.128: icmp_seq=3 ttl=64 time=0.957 ms
64 bytes from 192.168.133.128: icmp_seq=4 ttl=64 time=0.587 ms
64 bytes from 192.168.133.128: icmp_seq=5 ttl=64 time=0.724 ms
64 bytes from 192.168.133.128: icmp_seq=6 ttl=64 time=8.35 ms
64 bytes from 192.168.133.128: icmp_seq=7 ttl=64 time=1.03 ms
64 bytes from 192.168.133.128: icmp_seq=8 ttl=64 time=1.16 ms
64 bytes from 192.168.133.128: icmp_seq=9 ttl=64 time=1.07 ms
64 bytes from 192.168.133.128: icmp_seq=10 ttl=64 time=0.728 ms
64 bytes from 192.168.133.128: icmp_seq=11 ttl=64 time=0.716 ms

```

Figure 5. Connectivity test between Kali Linux and Metasploitable2.

3.3 Web Application Target

DVWA (Damn Vulnerable Web Application) was installed and accessible through the local web server. The application provides intentionally vulnerable modules for controlled web-security learning.

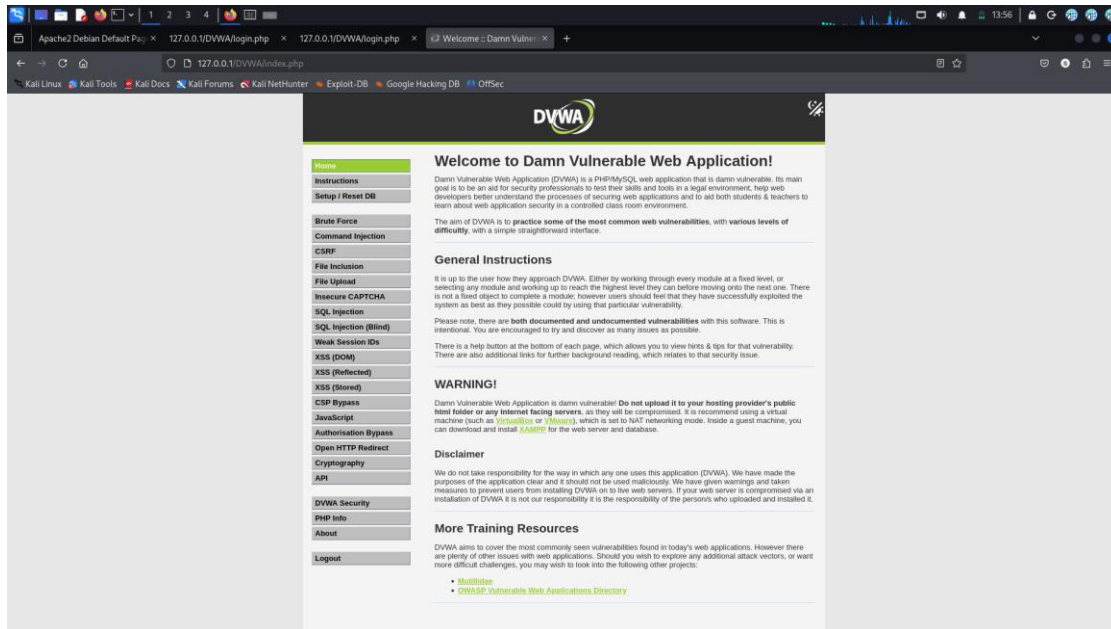


Figure 6. DVWA dashboard successfully loaded.

4. Linux Fundamentals

4.1 File System Navigation

The commands `pwd`, `ls`, and `cd` were practiced to identify the current working directory, list files, and navigate between directories.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)~  
$ pwd  
/home/kali  
(kali@kali)~  
$ ls  
AutoRecon          metasploitable_nmap.txt  reversehack.exe  
bindhack1.exe      Music                    rXmpBMHT.jpeg  
bindhack2.exe      network_tool            shell.php.jpg  
bindhack3.exe      output.txt              shell.phtml  
bindhck.exe        pentestlab              shortlist.txt  
Desktop            Pictures                 Sniper  
Documents          pJPqGdBe.jpeg          sqlmap-dev  
Downloads          Public                  Templates  
DVWA               RctShRDU.jpeg          Videos  
hash.txt           reversehack1.exe        VzOJmRCi.jpeg  
Hping3             reversehack2.exe        'Working From Home Information.txt'  
hydra.restore      reversehack3.exe        zphisher  
id_rsa             reversehack4.exe  
'Kali Linux 4k.jpg' reversehack5.exe  
(kali@kali)~  
$ ls -la  
total 151984  
drwx----- 33 kali kali    4096 Aug 14 14:34 .  
drwxr-xr-x  3 root root    4096 Nov 30 2024 ..  
drwxr-xr-x  3 root root    4096 Jul  4 2025 AutoRecon  
-rw-r--r--  1 kali kali     220 Nov 30 2024 .bash_logout  
-rw-r--r--  1 kali kali   5551 Nov 30 2024 .bashrc
```

Figure 7. Linux filesystem navigation.

4.2 File and Directory Permissions

Linux permissions were examined and modified using commands such as `chmod` and `chown`. The exercise demonstrated ownership and read/write/execute permission concepts.

```
(kali㉿kali)-[~]
$ cd ~

(kali㉿kali)-[~]
$ echo "Cybersecurity Lab Test" > lab_test.txt

(kali㉿kali)-[~]
$ ls -l lab_test.txt
-rw-rw-r-- 1 kali kali 23 Aug 14 14:45 lab_test.txt

(kali㉿kali)-[~]
$ chmod 600 lab_test.txt

(kali㉿kali)-[~]
$ ls -l lab_test.txt
-rw----- 1 kali kali 23 Aug 14 14:45 lab_test.txt

(kali㉿kali)-[~]
$
```

Figure 8. Linux file permissions and ownership.

4.3 Package Management

APT and dpkg were used for package installation, updates, and package configuration in Kali Linux.

```
(kali㉿kali)-[~]
$ apt policy nmap
nmap:
  Installed: 7.95+dfsg-1kali1
  Candidate: 7.99+dfsg-1kali1
  Version table:
     7.99+dfsg-1kali1 500
        500 https://http.kali.org/kali kali-rolling/non-free amd64 Packages
  ** 7.95+dfsg-1kali1 100
        100 /var/lib/dpkg/status

(kali㉿kali)-[~]
$ dpkg -s nmap | head
Package: nmap
Status: install ok installed
Priority: optional
Section: non-free/net
Installed-Size: 4580
Maintainer: Kali Developers <devel@kali.org>
Architecture: amd64
Version: 7.95+dfsg-1kali1
Depends: nmap-common (= 7.95+dfsg-1kali1), libc6 (≥ 2.38), libgcc-s1 (≥ 3.0), liblinear4 (≥ 2.01+dfsg), liblu
a5.4-0 (≥ 5.4.6), libpcap0.8t64 (≥ 1.5.1), libpcr2-8-0 (≥ 10.32), libssh2-1t64 (≥ 1.11.0), libssl3t64 (≥ 3
.0.0), libstdc++6 (≥ 11), zlib1g (≥ 1:1.1.4)
Suggests: ncat, ndiff, zenmap
```

Figure 9. Package management using apt/dpkg.

4.4 Networking Commands

Commands such as ip/ifconfig, ping, netstat/ss, and traceroute were practiced to inspect interfaces, connectivity, routes, and listening services.

```
└─$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 127.0.0.1:3306          0.0.0.0:*               LISTEN
tcp6       0      0 127.0.0.1:43855         :::*                    LISTEN
tcp6       0      0 127.0.0.1:8080          :::*                    LISTEN
tcp6       0      0 :::80                   :::*                    LISTEN
udp        0      0 0.0.0.0:36387           0.0.0.0:*
udp        0      0 192.168.133.129:3702    0.0.0.0:*
udp        0      0 239.255.255.250:3702    0.0.0.0:*
udp        0      0 192.168.241.130:3702    0.0.0.0:*
udp        0      0 239.255.255.250:3702    0.0.0.0:*
udp        0      0 0.0.0.0:58752           0.0.0.0:*
udp6       0      0 fe80::5e8:c61d:4ad:3702 :::*
udp6       0      0 ff02::c:3702            :::*
udp6       0      0 fe80::1df2:289b:c3:3702 :::*
udp6       0      0 ff02::c:3702            :::*
udp6       0      0 :::38646                 :::*
udp6       0      0 :::57280                 :::*

└─(kali㉿kali)-[~]
└─$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1 192.168.241.2 (192.168.241.2) 10.285 ms 10.050 ms 9.956 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
```

Figure 10. Linux networking commands.

5. Networking Basics

5.1 OSI Model

Layer	Name	Main Function	Examples
7	Application	Provides services to applications	HTTP, DNS

6	Presentation	Data representation, encoding, encryption	TLS, encoding
5	Session	Establishes and manages sessions	Session control
4	Transport	End-to-end delivery and ports	TCP, UDP
3	Network	Logical addressing and routing	IPv4, ICMP
2	Data Link	Local frame delivery	Ethernet, ARP
1	Physical	Transmits raw bits	Cable, radio

5.2 TCP/IP Protocol Suite

Layer	Function	Examples
Application	Network services for applications	HTTP, DNS, FTP
Transport	End-to-end communication and port delivery	TCP, UDP
Internet	Addressing and routing	IPv4, ICMP
Network Access	Local network delivery	Ethernet, ARP

5.3 IP Addressing, Subnetting and NAT

The Host-only network uses 192.168.133.0/24 with subnet mask 255.255.255.0. The usable host range is 192.168.133.1–192.168.133.254 and the broadcast address is 192.168.133.255. Kali's NAT interface provides a separate path for Internet access.

```

(kali㉿kali)-[~]
$ ip addr show eth1
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:03:0e:e8 brd ff:ff:ff:ff:ff:ff
    inet 192.168.133.129/24 brd 192.168.133.255 scope global dynamic noprefixroute eth1
        valid_lft 1752sec preferred_lft 1752sec
    inet6 fe80::5e8:c61d:4ad0:1008/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali㉿kali)-[~]
$ ip route
default via 192.168.241.2 dev eth0 proto dhcp src 192.168.241.130 metric 102
192.168.133.0/24 dev eth1 proto kernel scope link src 192.168.133.129 metric 101
192.168.241.0/24 dev eth0 proto kernel scope link src 192.168.241.130 metric 102

(kali㉿kali)-[~]
$ ping -c 4 192.168.133.128
PING 192.168.133.128 (192.168.133.128) 56(84) bytes of data:
64 bytes from 192.168.133.128: icmp_seq=1 ttl=64 time=4.47 ms
64 bytes from 192.168.133.128: icmp_seq=2 ttl=64 time=0.694 ms
64 bytes from 192.168.133.128: icmp_seq=3 ttl=64 time=0.615 ms
64 bytes from 192.168.133.128: icmp_seq=4 ttl=64 time=0.815 ms

— 192.168.133.128 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3041ms
rtt min/avg/max/mdev = 0.615/1.647/4.466/1.628 ms

(kali㉿kali)-[~]
$

```

Figure 11. IP addressing and routing information.

5.4 DNS

DNS converts domain names into IP addresses. The lab verified resolver configuration and domain resolution using command-line tools.

```

(kali㉿kali)-[~]
$ cat /etc/resolv.conf
# Generated by NetworkManager
search localdomain
nameserver 192.168.241.2
nameserver 192.168.133.1

(kali㉿kali)-[~]
$ nslookup google.com
Server:      192.168.241.2
Address:     192.168.241.2#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.193.14
Name:   google.com
Address: 2404:6800:4013:813::65
Name:   google.com
Address: 2404:6800:4013:813::71
Name:   google.com
Address: 2404:6800:4013:813::66
Name:   google.com
Address: 2404:6800:4013:813::8a

(kali㉿kali)-[~]
$ ping -c 4 google.com
PING google.com (142.250.193.14) 56(84) bytes of data:
64 bytes from tzelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=1 ttl=128 time=73.8 ms
64 bytes from tzelb-bg-in-f14.1e100.net (142.250.193.14): icmp_seq=2 ttl=128 time=67.0 ms
^C

```

Figure 12. DNS practical test.

```

(kali㉿kali)-[~]
$ nslookup google.com
Server:      192.168.241.2
Address:     192.168.241.2#53

Non-authoritative answer:
Name:   google.com
Address: 192.178.177.101
Name:   google.com
Address: 192.178.177.138
Name:   google.com
Address: 192.178.177.102
Name:   google.com
Address: 192.178.177.113
Name:   google.com
Address: 192.178.177.139
Name:   google.com
Address: 192.178.177.100
Name:   google.com
Address: 2404:6800:4002:831::200e

(kali㉿kali)-[~]
$ ping -c 4 google.com
PING google.com (192.178.177.100) 56(84) bytes of data.
64 bytes from lcdclw-in-f100.1e100.net (192.178.177.100): icmp_seq=1 ttl=128 time=53.6 ms
64 bytes from lcdclw-in-f100.1e100.net (192.178.177.100): icmp_seq=2 ttl=128 time=48.4 ms

```

Figure 13. DNS resolution output.

5.5 HTTP/HTTPS

HTTP connectivity to the target web server was tested with curl. The practical work also distinguished HTTP from HTTPS and examined TLS-related behavior.

```

(kali㉿kali)-[~]
$ curl -I http://192.168.133.128
HTTP/1.1 200 OK
Date: Sat, 15 Aug 2026 12:20:07 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

```

Figure 14. HTTP practical test.

```
(kali㉿kali)-[~]
└─$ curl -I http://192.168.133.128
HTTP/1.1 200 OK
Date: Fri, 14 Aug 2026 18:59:29 GMT
Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html
```

Figure 15. HTTP response from the target web server.

6. Cryptography Basics

6.1 Symmetric vs Asymmetric Encryption

Feature	Symmetric	Asymmetric
Keys	One shared secret key	Public/private key pair
Speed	Generally faster	Generally slower
Examples	AES	RSA, ECC
Typical use	Bulk data encryption	Key exchange, authentication, digital signatures

6.2 OpenSSL Encryption and Decryption

A test file/message was encrypted with OpenSSL using AES and then decrypted. The exercise demonstrated that the original content can be recovered when the correct decryption key/password is available.

```
(kali㉿kali)-[~/crypto-lab]
$ cat decrypted.txt
This is my cybersecurity cryptography lab.

(kali㉿kali)-[~/crypto-lab]
$ diff message.txt decrypted.txt

(kali㉿kali)-[~/crypto-lab]
$ sha256sum message.txt decrypted.txt
053be2a7a418f5f31a94a8546b065ed686bda9f9d50d55707369eab29ac207b8 message.txt
053be2a7a418f5f31a94a8546b065ed686bda9f9d50d55707369eab29ac207b8 decrypted.txt

(kali㉿kali)-[~/crypto-lab]
$
```

Figure 16. OpenSSL decryption operation and recovered content.

```
Session Actions Edit View Help
kali㉿kali: ~/crypto-lab

(kali㉿kali)-[~]
$ openssl version
OpenSSL 3.6.3 9 Jun 2026 (Library: OpenSSL 3.6.3 9 Jun 2026)

(kali㉿kali)-[~]
$ mkdir -p ~/crypto-lab
cd ~/crypto-lab

(kali㉿kali)-[~/crypto-lab]
$ pwd
/home/kali/crypto-lab

(kali㉿kali)-[~/crypto-lab]
$ echo "This is my cybersecurity cryptography lab." > message.txt

(kali㉿kali)-[~/crypto-lab]
$ cat message.txt
This is my cybersecurity cryptography lab.

(kali㉿kali)-[~/crypto-lab]
$ openssl enc -aes-256-cbc -salt -pbkdf2 -in message.txt -out encrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:

(kali㉿kali)-[~/crypto-lab]
$ ls -l
total 8
-rw-rw-r-- 1 kali kali 64 Aug 15 07:36 encrypted.bin
-rw-rw-r-- 1 kali kali 43 Aug 15 07:35 message.txt

(kali㉿kali)-[~/crypto-lab]
```

Figure 17. Additional cryptography/OpenSSL practical evidence.

6.3 Hashing

MD5 and SHA-256 hashes were calculated. Hashing is one-way and is primarily used for integrity verification rather than reversible encryption.

```

(kali@kali)-[~/crypto-lab]
$ md5sum message.txt
d305b37508af43016c8ca865b414957e message.txt

(kali@kali)-[~/crypto-lab]
$ sha256sum message.txt
053be2a7a418f5f31a94a8546b065ed686bda9f9d50d55707369eab29ac207b8 message.txt

(kali@kali)-[~/crypto-lab]
$ cp message.txt message_modified.txt

(kali@kali)-[~/crypto-lab]
$ echo "Modified" >> message_modified.txt

(kali@kali)-[~/crypto-lab]
$ sha256sum message.txt message_modified.txt
053be2a7a418f5f31a94a8546b065ed686bda9f9d50d55707369eab29ac207b8 message.txt
92a0247285b5fb64c50ca046ebb61e1091a73a852e644bdba3bb65fe90291369 message_modified.txt

(kali@kali)-[~/crypto-lab]
$

```

Figure 18. MD5 and SHA-256 hashing practical.

6.4 Digital Certificates

A 2048-bit RSA key and an X.509 certificate were generated with OpenSSL. Certificate fields such as subject, issuer, and validity were inspected.

```

kali@kali: ~/crypto-lab
Session Actions Edit View Help
$ openssl x509 -in server.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      13:c1:7a:f6:61:a2:e6:d9:3c:56:5c:84:62:f6:ab:8b:7a:66:70:47
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
    Validity
      Not Before: Aug 15 11:49:01 2026 GMT
      Not After : Aug 15 11:49:01 2027 GMT
    Subject: C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:d1:8c:6e:dc:48:47:fc:80:90:b5:51:8a:a7:45:
        7f:dc:b0:46:8b:dc:6d:08:29:c6:87:04:89:de:3a:
        41:28:2d:c2:a8:ee:78:47:04:5f:73:fb:95:bb:5e:
        e0:60:43:48:90:a5:8a:4b:22:65:6f:4a:08:ec:9d:
        61:b2:3b:b7:23:bd:d9:76:e9:04:74:59:2c:ff:8c:
        c0:c0:bb:f4:e3:f5:83:c8:fc:46:df:0e:d7:c2:33:
        d1:b6:d1:8e:23:0e:3f:90:81:fe:2a:bc:43:a9:4e:
        f0:8e:b7:5c:67:ec:51:ff:04:bd:a0:83:d6:d2:b5:
        23:27:45:69:59:06:23:39:fc:d7:2a:55:cd:d5:b4:
        21:ec:b8:e5:fe:dd:b0:e7:8f:b6:46:b4:67:cf:56:
        7d:a8:47:c2:6a:bc:d5:4f:63:2c:51:b7:01:71:42:
        50:12:9d:73:ca:b7:e1:bb:53:38:1a:0e:8b:43:b2:
        55:0b:41:1c:d8:8c:c9:59:7e:8e:26:da:41:3b:ea:
        28:d5:63:34:55:76:c4:14:55:01:a1:b7:70:14:4a:
        c5:80:91:e9:d2:ad:d0:8c:6e:37:75:e0:8c:3e:43:
        79:65:3d:fd:81:56:7c:58:48:5e:b2:49:e1:f6:7e:
        d4:84:10:c5:0d:87:80:6e:58:bc:73:13:df:42:f6:
        c0:4b
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Subject Key Identifier:
        E6:4D:A9:1B:5C:AC:9C:3B:1C:DF:EC:F8:D4:21:4F:42:7C:34:5C:1D

```

Figure 19. X.509 digital certificate details.

6.5 SSL/TLS

A local TLS test environment was configured using the generated certificate and private key. The exercise demonstrated certificate-based encrypted communication.

```
(kali㉿kali)-[~]
$ openssl s_client -connect 127.0.0.1:8443
Connecting to 127.0.0.1
CONNECTED(00000003)
Can't use SSL_get_servername
depth=0 C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
verify error:num=18:self-signed certificate
verify return:1
depth=0 C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
verify return:1
---
Certificate chain
 0 s:C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
  i:C=IN, ST=WB, L=KOLKATA, O=CYBERSECURITY LAB, OU=CYBER, CN=LOCALHOST
  a:PKEY: RSA, 2048 (bit); sigalg: sha256WithRSAEncryption
  v:NotBefore: Aug 15 11:49:01 2026 GMT; NotAfter: Aug 15 11:49:01 2027 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIDuTCCAqGgAwIBAgIUe8F69mGi5tk8VlyEYvari3pmcEcwDQYJKoZIhvcNAQEL
BQAwbDELMAkGA1UEBhMCSU4xCzAJBgNVBAGMAldCMRAwDgYDVQQHDAdLT0xLQVRB
MRowGAYDVQQKDBFduUjFvNFQ1VSSVRZIEExBQjEOMAwGA1UECwwFQ1lCRVixEjAQ
BgNVBAMMCUxPQ0FMSE9TVDAeFw0yNjA4MTUxMTQ5MDFaFw0yNzA4MTUxMTQ5MDFa
MGwxCzAJBgNVBAYTAkxMQswCQYDVQQIDAJXQjEQA4GA1UEBwwHS09MS0FUQTEa
MBGGA1UECgwrQ1lCRVJTRUNVUklUWSBMQUIxOjA4MTUxMTQ5MDFaFw0yNzA4MTUxMTQ5MDFa
VQDDALMT0NBTEhPU1QwggEiMA0GCSqGSIb3DQEBAQUAA4IBDwAwggEKAoIBAQR
jG7cSEf8gJC1UYqnRX/csEaL3G0IKcaHBIne0kEoLcKo7nhHBF9z+5W7XuBgQ0iQ
pYpLImVvSgjsnWgy07cjvdl26QR0WSz/jMDAu/Tj9YPI/EbfDtFCM9G20Y4jDj+Q
gf4qvE0pTvC0t1xn7FH/BL2gg9bStSmnRWLZBiM5/NcqVc3VtCHsu0X+3bDnj7ZG
tGfPVn2oR8JqvNVPYyxRtwFxQlASnXPKt+G7UzgaDotDsLULQRzYjMlZfo4m2kE7
```

Figure 20. SSL/TLS practical evidence.

7. Tool Familiarization

7.1 Nmap

Nmap was used to identify reachable ports and services on the intentionally vulnerable target in the isolated lab.

```
kali@kali: /var/www/html/DVWA/config
$ nmap 192.168.133.128
Starting Nmap 7.95 ( https://nmap.org ) at 2026-08-14 14:04 EDT
Nmap scan report for 192.168.133.128
Host is up (0.00079s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:4B:CF:8D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 6.92 seconds
```

Figure 21. Nmap scan results.

7.2 Wireshark

Wireshark was used to capture and inspect network packets. Both HTTP and ICMP traffic were observed to understand how protocol data appears at packet level.

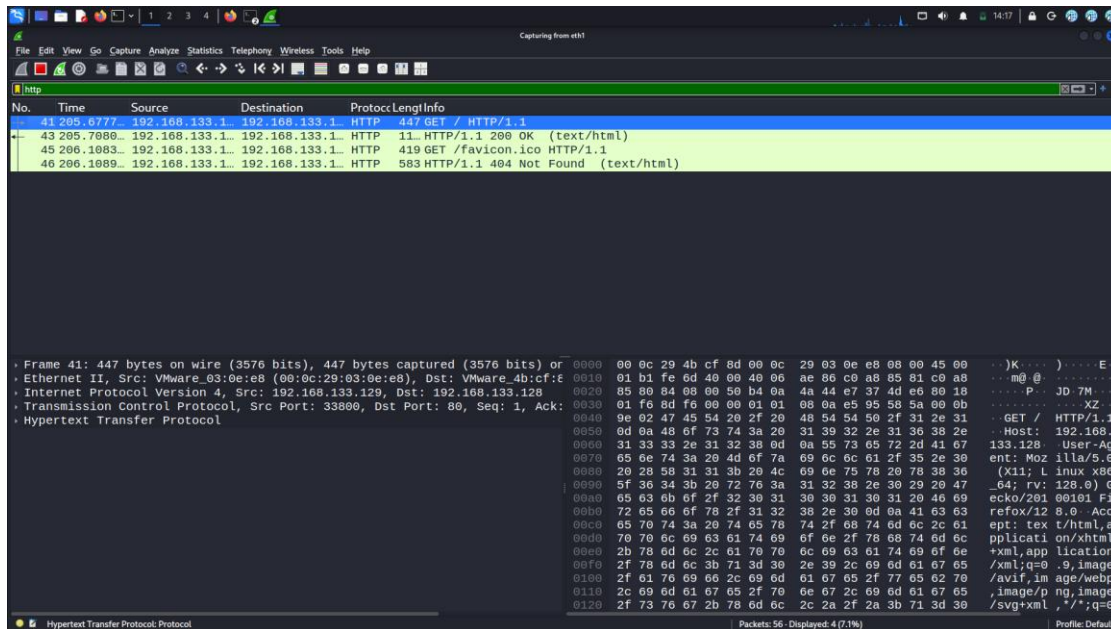


Figure 22. Wireshark HTTP packet capture.

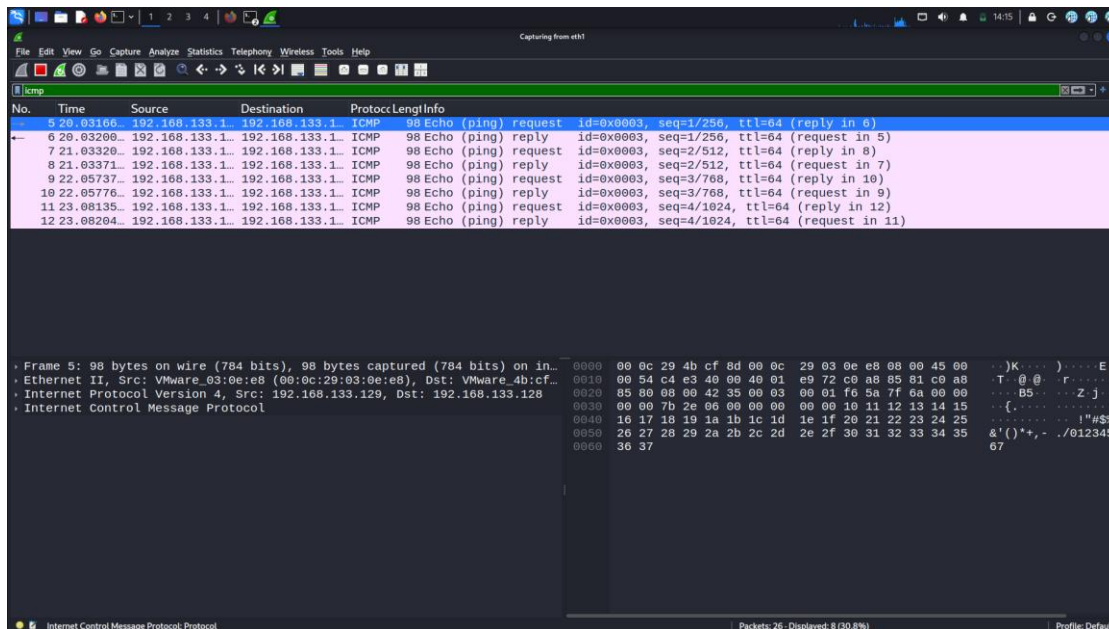


Figure 23. Wireshark ICMP packet capture.

7.3 Burp Suite

Burp Suite was configured as a web proxy and used to observe HTTP requests generated while interacting with the DVWA environment.

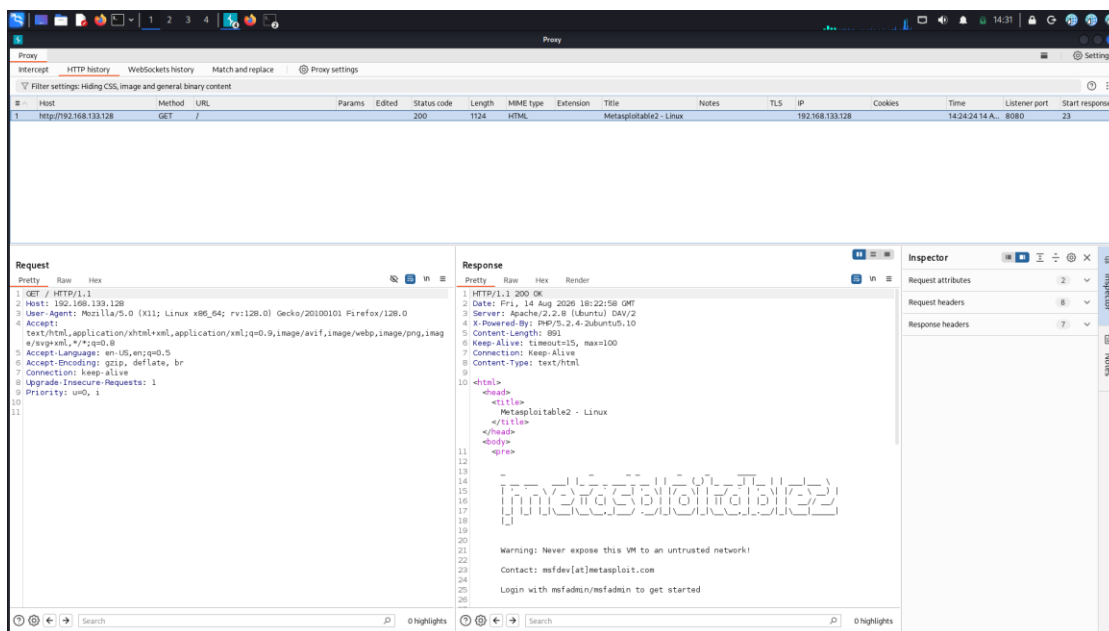


Figure 24. Burp Suite web traffic/proxy evidence.

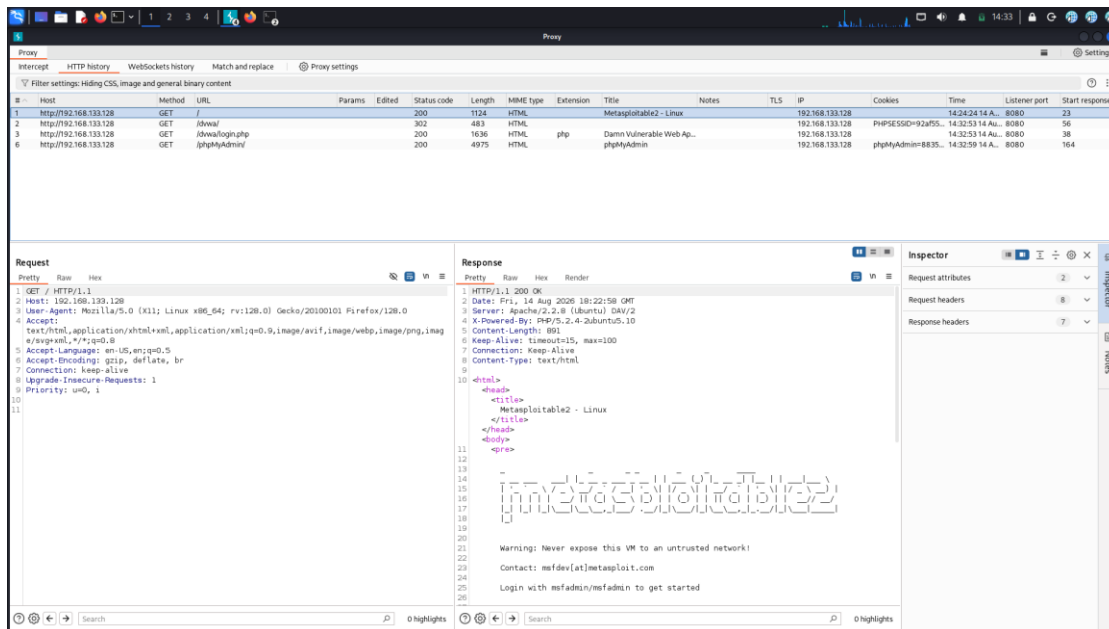


Figure 25. DVWA request/interaction evidence.

7.4 Netcat

Netcat was used for a basic TCP listener/client test, demonstrating simple network connectivity and troubleshooting.

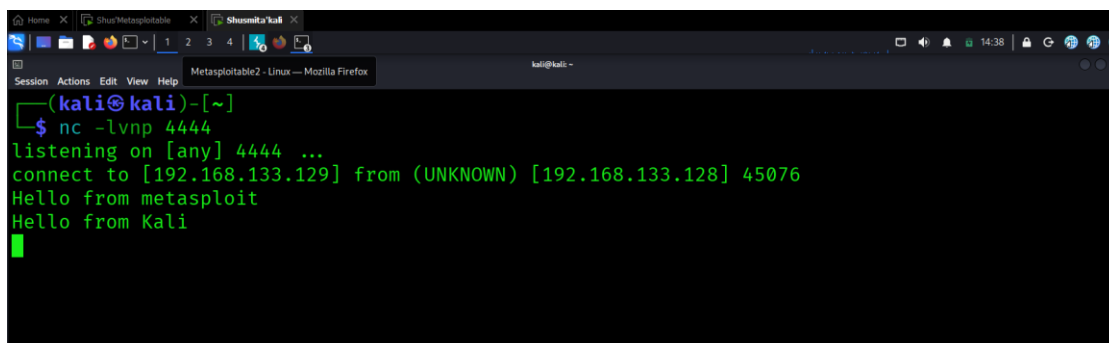


Figure 26. Netcat listener/client test on Kali Linux.

```

TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:21437 (20.9 KB) TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ ping -c 192.168.133.128
Usage: ping [-LRUbdfnqrvaA] [-c count] [-i interval] [-w deadline]
          [-p pattern] [-s packetsize] [-t ttl] [-I interface or address]
          [-M mtu discovery hint] [-S sndbuf]
          [-T timestamp option] [-Q tos] [hop1 ...] destination
msfadmin@metasploitable:~$ ping -c 4 192.168.133.128
PING 192.168.133.128 (192.168.133.128) 56(84) bytes of data.
64 bytes from 192.168.133.128: icmp_seq=1 ttl=64 time=1.55 ms
64 bytes from 192.168.133.128: icmp_seq=2 ttl=64 time=0.073 ms
64 bytes from 192.168.133.128: icmp_seq=3 ttl=64 time=0.053 ms
64 bytes from 192.168.133.128: icmp_seq=4 ttl=64 time=0.071 ms

--- 192.168.133.128 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2999ms
rtt min/avg/max/mdev = 0.053/0.436/1.550/0.643 ms
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ nc 192.168.133.129 4444
Hello from metasploit
Hello from Kali

```

Figure 27. Netcat test from the target-side environment.

8. Observations and Results

- The virtual machines successfully communicated through the isolated Host-only network.
- Kali Linux successfully maintained NAT and Host-only connectivity.
- Apache and DVWA were accessible and operational in the lab.
- Linux navigation, permissions, package management, and networking commands were successfully practiced.
- DNS resolution and HTTP communication were verified.
- OpenSSL exercises demonstrated encryption/decryption, hashing, certificate generation, and TLS concepts.
- Nmap, Wireshark, Burp Suite, and Netcat were successfully used for basic network and web-security analysis.

9. Conclusion

The cybersecurity lab was successfully established as a controlled environment for authorized practical learning. The project covered cybersecurity fundamentals, Linux administration, networking, cryptography, and security-tool familiarization. The combination of an isolated Host-only network, Kali Linux, and intentionally vulnerable targets provides a suitable foundation for further penetration-testing and defensive-security exercises.

10. References

- Kali Linux documentation and manual pages.
- OpenSSL documentation and manual pages.
- Nmap reference documentation.
- Wireshark documentation.
- Burp Suite documentation.
- DVWA project documentation.
- Metasploitable2 lab documentation.